Summer Research Synopsis

A Graph Edit Metric for Synthetic Data Utility

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Summary

1. Background

What are we working with, the issue we face, and a what has been done.

2. Ideas

Previous ideas, trial and error...

3. Proposed Method

Our method for solving the problem, how it works.

4. Results

Experiments and results compared to contemporary utility methods.

5. Future Work

Limitations, improvements, future applications.

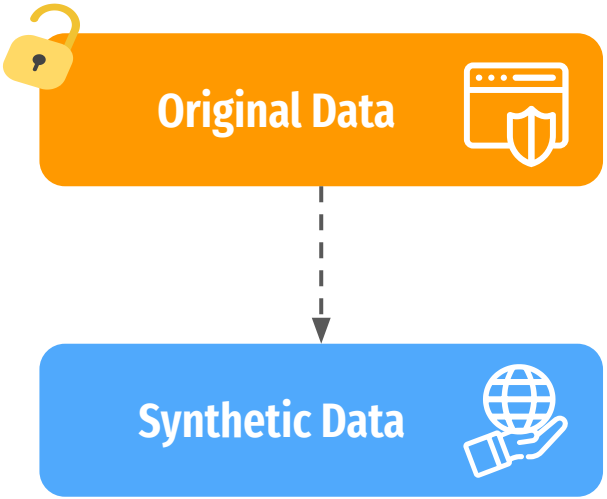
1. Background

Sharing Data in a Private World

As artificial Intelligence models are being used in more sensitive fields, researchers struggle to dissemination data due to growing privacy restrictions.

To continue scientific development, **researchers must find a way to disseminate data while adhering to privacy regulations**.

Necessity for Synthetic Data Generators.



Outsiders will use synthetic data…

Was their time wasted?

Compared to the original data…

- Would they come to the same conclusions?
- Would their model perform the same?
- Etc…

3 Dimensions of Synthetic Data Quality Evaluation

Lots of Focus

Privacy

- Does the synthetic data not jeopardize the privacy of the original dataset?

Fidelity

- Is the synthetic data statistical similarity to the original dataset?

Utility

- Does the synthetic data perform comparably to the original data for the same task?

*aws.amazon.com/blogs/machine-learning/how-to-evaluate-the-quality-of-the-synthetic-data-measuring-from-the-perspective-of-fidelity-utility-and-privacy/

*Hittmeir, M., Ekelhart, A., & Mayer, R. (2019, August). On the utility of synthetic data: An empirical evaluation on machine learning tasks. In Proceedings of the 14th International Conference on Availability, Reliability and Security (pp. 1-6).

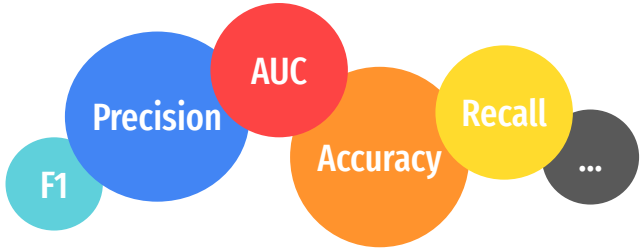
*Figueira, A., & Vaz, B. (2022). Survey on synthetic data generation, evaluation methods and GANs. Mathematics, 10(15), 2733.

Hu, B., Basri, M. A., Abdullah, A. Y. M., Tsao, S. F., Butt, Z., & Chen, H. (2023). Evaluation Methods for Synthetic Data in Pursuit of Open Data. Journal of Computational Vision and Imaging Systems, 9(1), 30-33.

Literature & Use Case

Current literature focuses on comparing model performance. This provides valuable insight on how the data performs for the task, but gloss over the effects on the innerworkings of the models the datasets are trained on.

Not many model structure focused metrics for Utility.



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*Figueira, A., & Vaz, B. (2022). **Survey on synthetic data generation, evaluation methods and GANs.** Mathematics, 10(15), 2733.

*Alaa, A., Van Breugel, B., Saveliev, E. S., & van der Schaar, M. (2022, June). **How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models**. In International Conference on Machine Learning (pp. 290-306). PMLR.

*Dankar, F. K., Ibrahim, M. K., & Ismail, L. (2022). **A multi-dimensional evaluation of synthetic data generators.** IEEE Access, 10, 11147-11158.

*Wang, L., Li, Q., Yu, Y., & Liu, J. (2018). **Region compatibility based stability assessment for decision trees.** *Expert Systems with Applications*, *105*, 112-128.

Jacobsen, B. N. (2023). **Machine learning and the politics of synthetic data**. Big Data & Society, 10(1), 20539517221145372.

2. Ideas

Ideas...

Original

```
pd.DataFrame(adult_res['original']).describe()
```

	accuracy	f1	auc	precision	recall
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	0.853788	0.848653	0.906075	0.847993	0.853788
std	0.005770	0.006089	0.007069	0.006294	0.005770
min	0.843688	0.837856	0.896131	0.836870	0.843688
25%	0.851946	0.846644	0.901470	0.845941	0.851946
50%	0.854379	0.849120	0.906081	0.848579	0.854379
75%	0.857774	0.852135	0.908644	0.852127	0.857774
max	0.861566	0.857146	0.919651	0.856543	0.861566

Synthetic

```
pd.DataFrame(adult_res['gc']).describe()
```

	accuracy	f1	auc	precision	recall
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	0.756159	0.674115	0.652095	0.705860	0.756159
std	0.002406	0.003528	0.009436	0.010823	0.002406
min	0.751519	0.670146	0.637186	0.685426	0.751519
25%	0.755010	0.672043	0.646607	0.700282	0.755010
50%	0.755961	0.673944	0.648725	0.704314	0.755961
75%	0.757480	0.674802	0.656955	0.713520	0.757480
max	0.760636	0.682915	0.667129	0.724469	0.760636

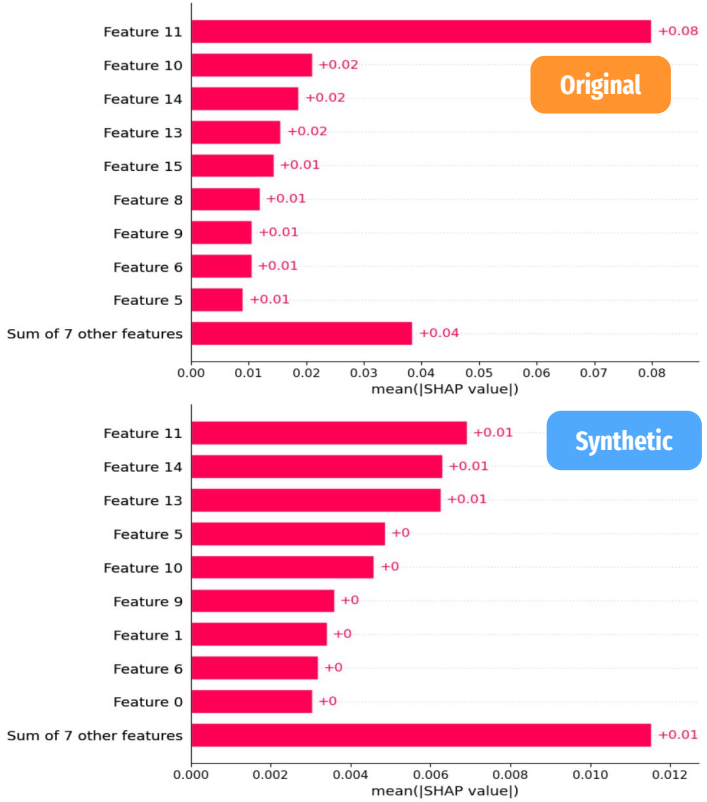
Cross-Validation Method

Hypothesis: the difference in volatility of model accuracy measures between datasets can determine the utility of synthetic data.

Outcome: differences model accuracy is a symptom of outliers and can not be used to distinguish original and synthetic data reliably.

Data generators generally decreased accuracy measure standard deviation because they struggle to create convincing outliers. Even though this trend exist, removing outliers in the original data causes original and synthetic datasets to become undistinguishable.

Ideas...



Feature Importance Method

Hypothesis: the difference in model feature importances between datasets can determine the utility of synthetic data.

Outcome: Data generators dim the importances of features which could be used to distinguish between original and synthetic data. How much it dims can be used to calculate a utility score.

This method already has a few papers talking about it with not much room for improvement, so we decided to move on.

Ideas...

```
print("Dataset: adult")
for key, value in results_adult.items():
    print(key, value)
```

Original

Synthetic ↓

```
Dataset: adult
orginal {'n_estimators': 388, 'criterion': 'log_loss', 'max_depth': 848, '
ctgan {'n_estimators': 951, 'criterion': 'entropy', 'max_depth': 647, 'mir
gc {'n_estimators': 841, 'criterion': 'log_loss', 'max_depth': 10, 'min_sa
pate {'n_estimators': 780, 'criterion': 'entropy', 'max_depth': 11, 'min_s
pb {'n_estimators': 989, 'criterion': 'gini', 'max_depth': 107, 'min_sampl
```

```
print("Dataset: iris")
for key, value in results_iris.items():
    print(key, value)
```

Original

Synthetic ↓

```
Dataset: iris
orginal {'n_estimators': 334, 'criterion': 'gini', 'max_depth': 922, 'min_
ctgan {'n_estimators': 13, 'criterion': 'log_loss', 'max_depth': 15, 'min_
gc {'n_estimators': 501, 'criterion': 'log_loss', 'max_depth': 966, 'min_s
pate {'n_estimators': 329, 'criterion': 'entropy', 'max_depth': 745, 'min_
pb {'n_estimators': 713, 'criterion': 'entropy', 'max_depth': 352, 'min_sa
```

*This example uses a random forest classifier, but you can use any classifier.

Hyperparameter Method

Hypothesis: a difference in model hyperparameter tuning between datasets can determine the utility of synthetic data.

Outcome: potentially a good heuristic, but unclear on how to calculate differences in parameter values and prone to subjectification.

Synthetic data generally showed greater numbers for numeric parameters, but justifying and quantifying the difference on a general scale is precarious.

3. Proposed Method

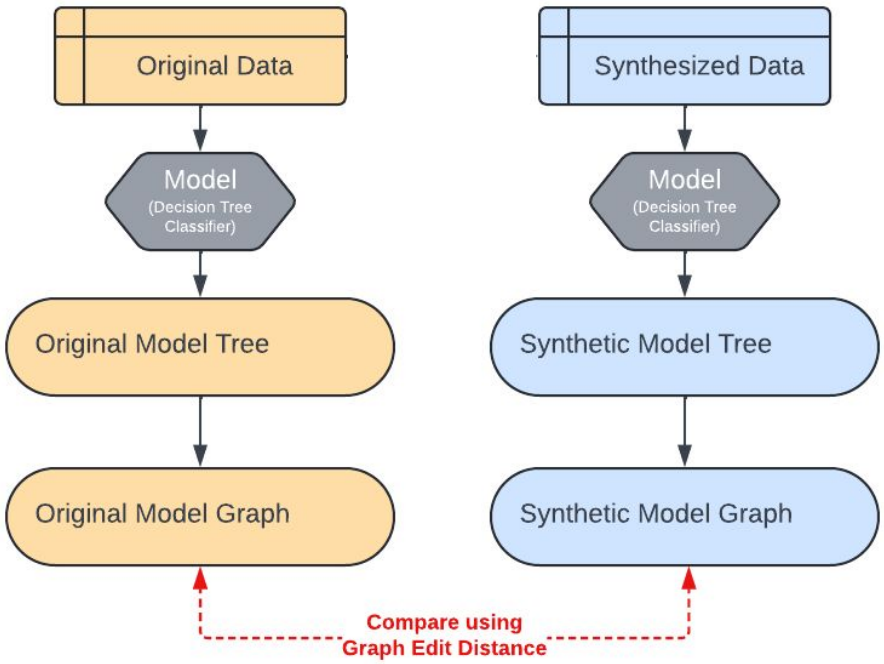
Proposed Method - Modified Graph Edit Distance

With a tree-based model, **we can use a modified graph edit distance to quantify the utility of synthetic data by comparing the difference the tree model structure between original and synthetic datasets.**

The higher our metric score, the closer the synthetic data performs to the original data.

Method Parts:

A) Tree to Graph Conversion
B) Graph Edit Algorithm

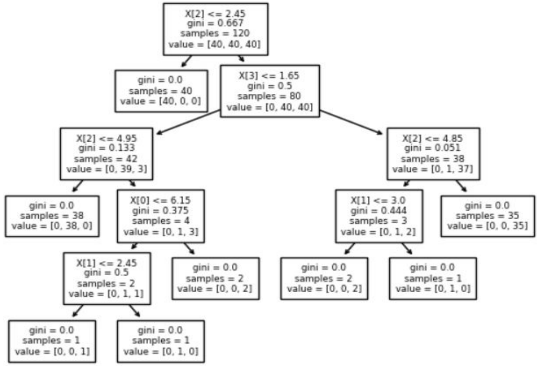


Why Graph Edit Distance instead of Tree Edit Distance

Tree Edit Distance (TED)

Expensive Metric

Resource heavy with exact algorithm capping $O(n^3)$ and approximations at $O(n^2 log(n))$ where n is the **amount of nodes in the tree**.



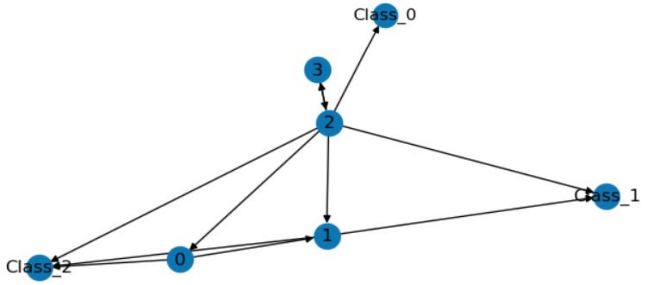
$O(15^3)$

$O(3375)$

Graph Edit Distance (GED)

Good Heuristic

Smaller information-dense representation that runs anywhere from $O(2^n)$, $O(n^3)$, or $O(n^2 log(n))$ depending on approximation where n is the **amount of features and classes in the dataset.**

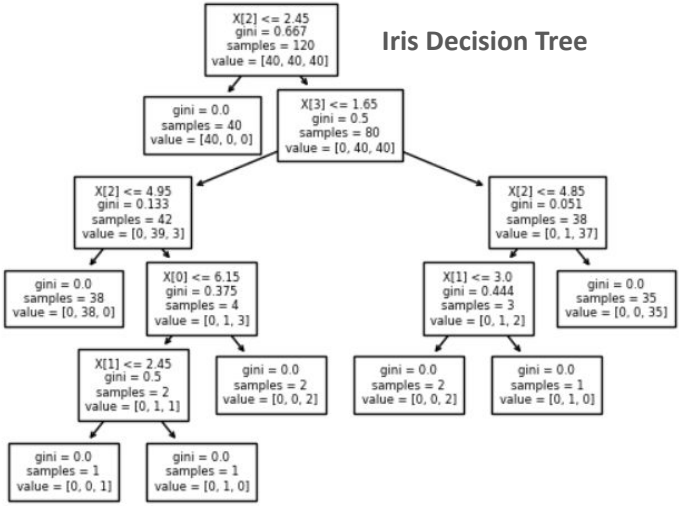


$O(2^7)$

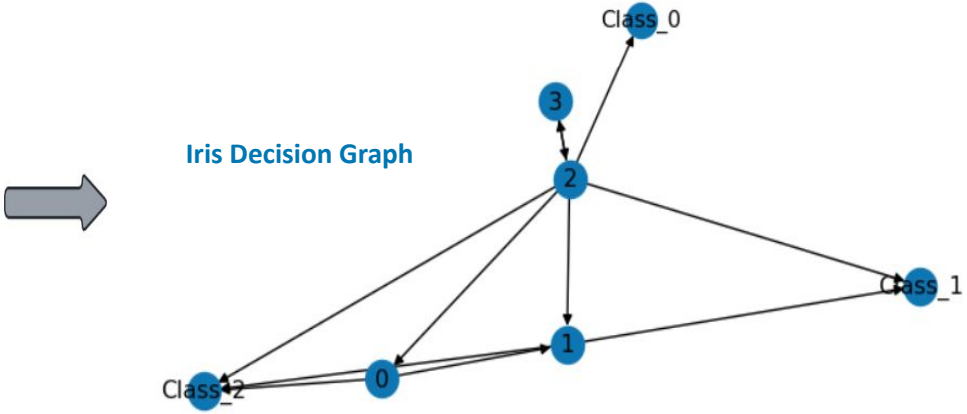
$O(128)$

Smallest example from the Iris dataset

Proposed Method - Tree to Graph



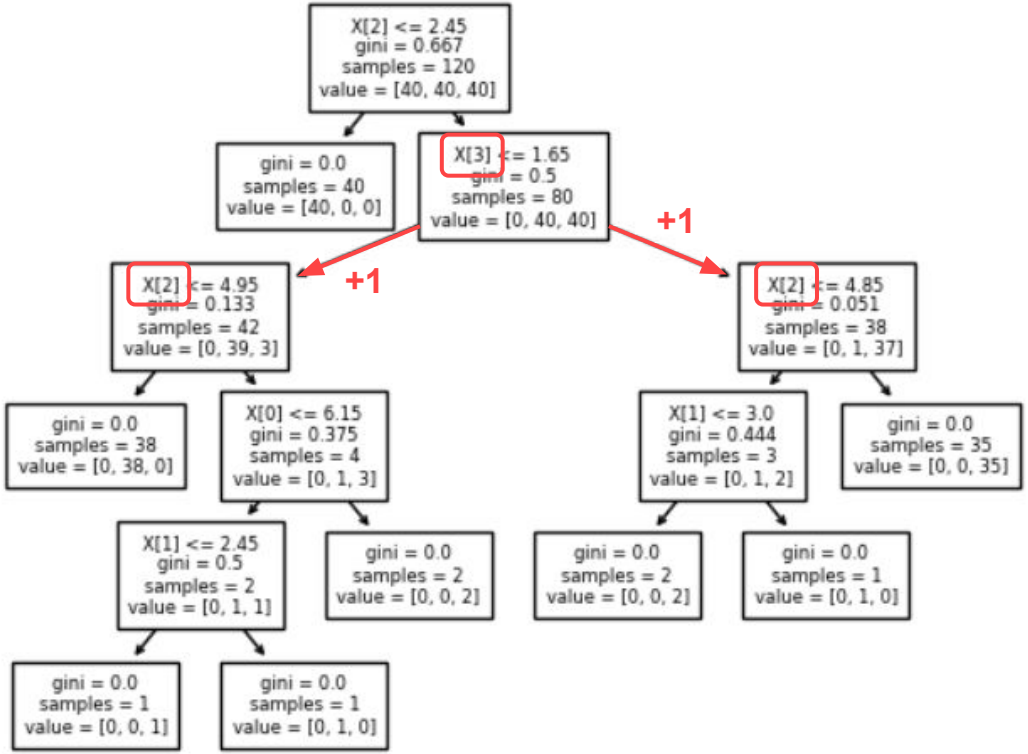
Iris Decision Tree



Iris Decision Graph

- Nodes represent features or classes.
- Nodes contain the average summary of the training samples that reached said node.
- Edges represent decision traversal.
- Edges contain the number of visits between said nodes.

Proposed Method - Tree to Graph (Edges)



Logic Traversal

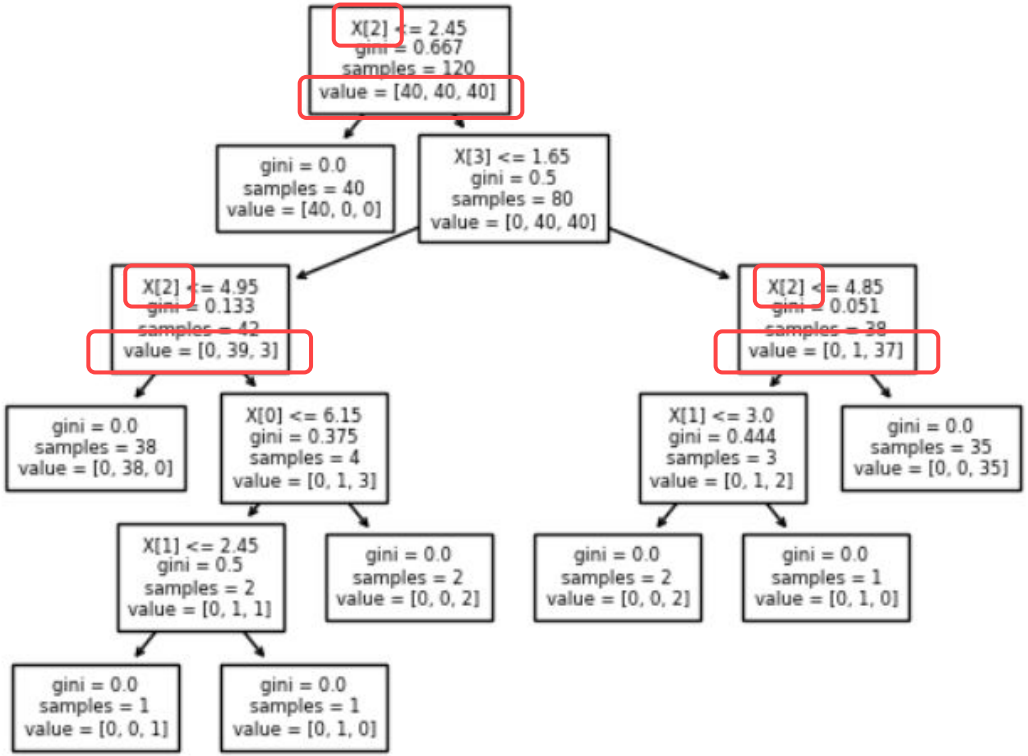
Iterates over each edge and records how many times a path is taken from node (feature/class) to node (feature/class).

Edge **[3][2]**
Weight = **2**

Edge **[2][3]**
Weight = **1**

Note: edges are directed and self loops are allowed.

Proposed Method - Tree to Graph (Nodes)



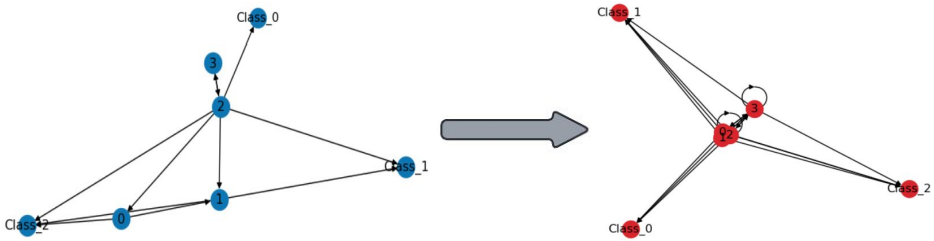
Output Traversal

Iterates over each node in the tree and records the average amount of observations (per class) that visit each unique node (feature/class).

Node [2]
Values = [[40, 40, 40], [0, 39, 3], [0, 1, 37]]
Avg = [13.33, 26.67, 26.67]

Proposed Method - Algorithm

Graph edit distance is a graph similarity measure analogous to Levenshtein distance for strings. It is defined as minimum cost of edit path (sequence of node and edge edit operations) transforming graph G1 to graph isomorphic to G2.



$$GED(g_1, g_2) = \min_{(e_1,\ldots,e_k)\in\mathcal{P}(g_1,g_2)} \sum_{i=1}^{k} c(e_i)$$

	L	e	v	e	n	s	h	t	e	i	n	
	L	e	v	e	n	s	h	t	e	i	n	
		L	**e**	**v**	**e**	**n**	**s**	**h**	**t**	**e**	**i**	**n**
	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5
L	0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
e	1	0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
v	1.5	1	0.5	0	0.5	1	1.5	2	2.5	3	3.5	4
e	2	1.5	1	0.5	0	0.5	1	1.5	2	2.5	3	3.5
n	2.5	2	1.5	1	0.5	0	0.5	1	1.5	2	2.5	3
s	3	2.5	2	1.5	1	0.5	0	0.5	1	1.5	2	2.5
h	3.5	3	2.5	2	1.5	1	0.5	0	0.5	1	1.5	2
t	4	3.5	3	2.5	2	1.5	1	0.5	0	0.5	1	1.5
e	4.5	4	3.5	3	2.5	2	1.5	1	0.5	0	0.5	1
i	5	4.5	4	3.5	3	2.5	2	1.5	1	0.5	0	0.5
n	5.5	5	4.5	4	3.5	3	2.5	2	1.5	1	0.5	0

Proposed Method - Algorithm (GED Score)

$$GED(g_1, g_2) = \min_{(e_1,\ldots,e_k)\in\mathcal{P}(g_1,g_2)} \sum_{i=1}^{k} c(e_i)$$

$$\Downarrow$$

$$GED_Score = 1 - \frac{GED(g_1, g_2)}{c_{node_del}|nodes_{g1}| + c_{edge_del}|edges_{g1}| + c_{node_ins}|nodes_{g2}| + c_{edge_ins}|edges_{g2}|}$$

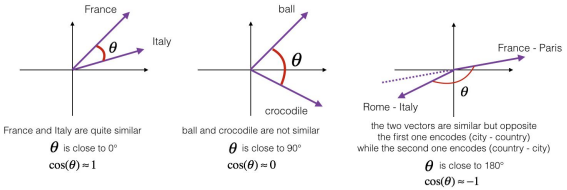
Provides a score between ***0 - 1*** given that the worst case is deleting and rebuilding all nodes/edges to make g_1 into g_2 where c_{node_del} is the cost to delete a node, c_{node_ins} is the cost to insert a node, c_{edge_del} is the cost to delete an edge, c_{edge_ins} is the cost to insert an edge.

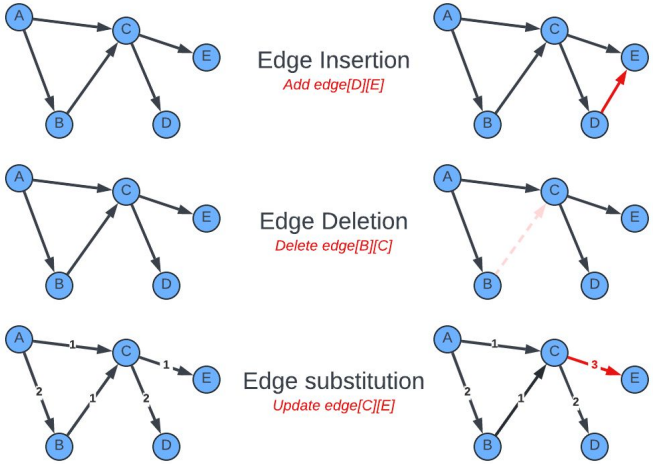
Assumes there isn't a negative cost for an operation.

Proposed Method - Algorithm (Costs)

Costs…

- Edge/Node Insertion: **1**
- Edge/Node Deletion: **1**
- Edge Substitution:
  - **0.5** if G_1[u][v].weight != G_2[u][v].weight
- Node Substitution:
  - **1 - cosine_sim**(G_1[x].avg, G_2[x].avg)





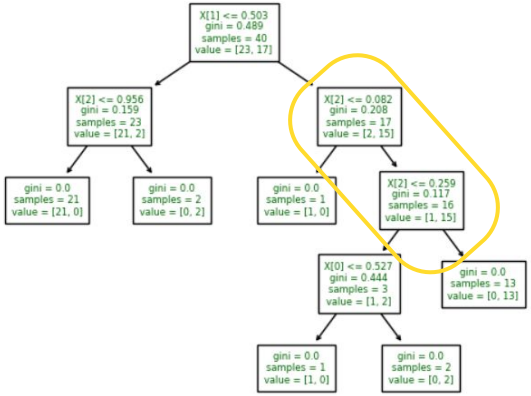
Edge [3][2]
Weight = 2

Node [2]
Avg = [13.33, 26.67, 26.67]

4. Results

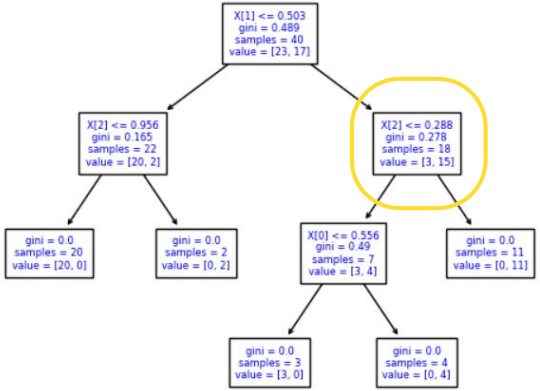
Simple Experiment

Dataset A - Tree



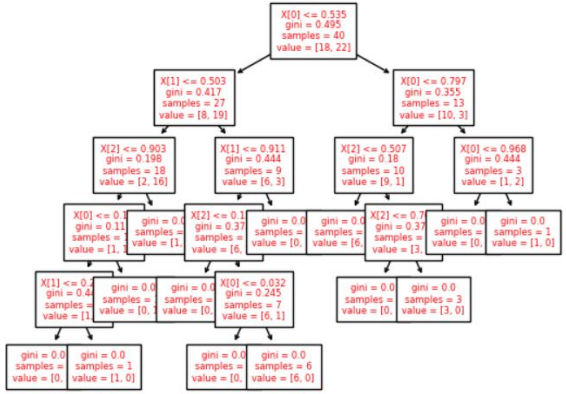
graph	accuracy	f1	auc
A	0.8	0.761905	0.750000

Dataset B - Tree



graph	accuracy	f1	auc
B	1.0	1.000000	1.000000

Dataset C - Tree



graph	accuracy	f1	auc
C	0.7	0.670330	0.666667

0.953 Sim Score

0.756 GED Sim Score

*Subsets of the same dataset

Data Setup

/ SynthData / data /

Name

- adult_original.csv
- bank_original.csv
- cancer_original.csv
- card_original.csv
- dermatology_original.csv
- diabetes_original.csv
- heart_original.csv
- iris_original.csv
- titanic_original.csv

Most Datasets from UCI

Diverse classification datasets

Datasets have been collected with minimal preprocessing.

```
Gaussian Copula
CTGAN
PATE-GAN
PrivBayes
```

Popular data synthesizes have been selected.

/ SynthData / data /

Name

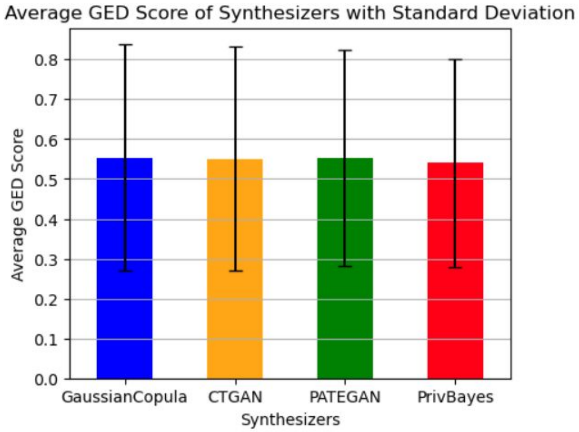
- cancer_gc.csv
- cancer_original.csv
- cancer_pate.csv
- cancer_pb.csv
- cancer_scores.csv
- card_ctgan.csv
- card_gc.csv
- card_original.csv
- card_pate.csv
- card_pb.csv
- card_scores.csv
- dermatology_ctgan.csv
- dermatology_gc.csv
- dermatology_original.csv
- dermatology_pate.csv
- dermatology_pb.csv
- dermatology_scores.csv

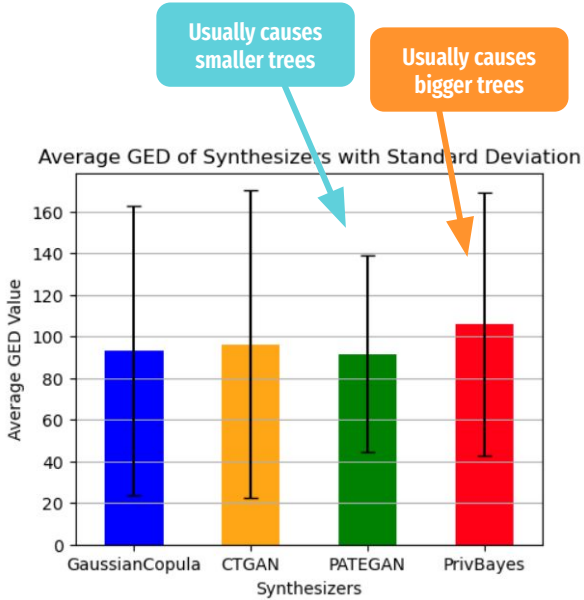
Datasets synthesized.

Results

On a macro level, our metric doesn't show a data synthesizer that performs better than the others.

On a micro level, you are able to distinguish and choose a "best" synthesizer for your dataset.

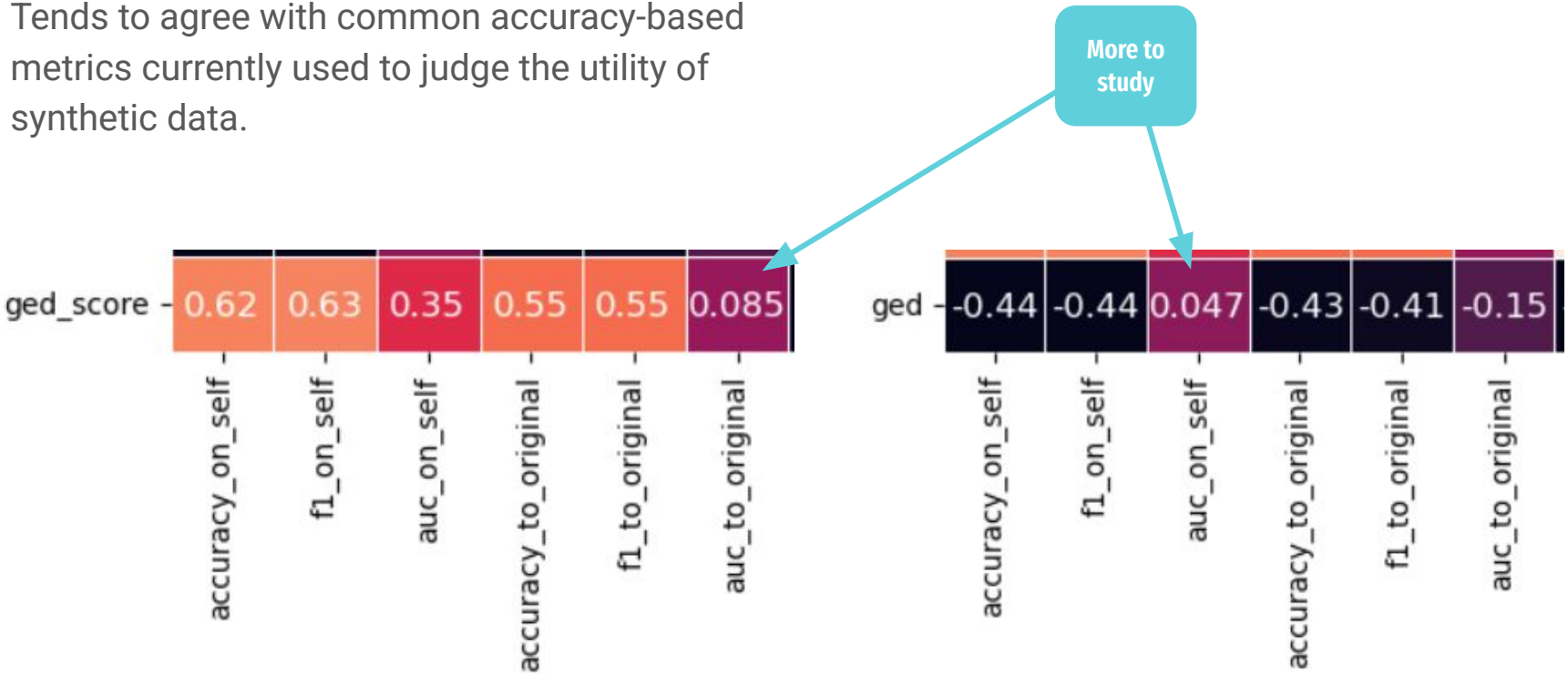




dataset	synthesizer	ged	ged_score
bank	original	0.0000	1.0000
bank	GaussianCopula	74.8749	0.7987
bank	CTGAN	78.7872	0.7710
bank	PATEGAN	88.6141	0.7668
bank	PrivBayes	84.0539	0.7872

Results

Tends to agree with common accuracy-based metrics currently used to judge the utility of synthetic data.



Results

But doesn't agree entirely with accuracy based metrics. This isn't bad, it just shows that is a different way to judge synthetic data utility.

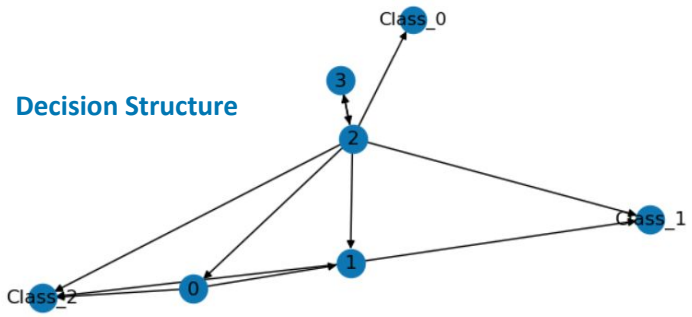
But having some similar correlation is a good sign that this metric could be used.

		Accuracy on Original	F1 on Original	AUC on Original	Raw GED	Proposed Metric
diabetes	original	0.7577	0.3944	0.5477	0.0000	1.0000
diabetes	GaussianCopula	0.7079	0.3309	0.4987	37.1287	0.9606
diabetes	CTGAN	0.7564	0.3850	0.5424	24.0583	0.9738
diabetes	PATEGAN	0.6958	0.3718	0.5309	60.1369	0.9317
diabetes	PrivBayes	0.7356	0.6011	0.7109	121.2478	0.8598
bank	original	0.8564	0.6610	0.6678	0.0000	1.0000
bank	GaussianCopula	0.7635	0.5225	0.5281	74.8749	0.7987
bank	CTGAN	0.8232	0.5986	0.5971	78.7872	0.7710
bank	PATEGAN	0.7436	0.5070	0.5087	88.6141	0.7668
bank	PrivBayes	0.5558	0.5558	0.5558	84.0539	0.7872
dermatology	original	0.9444	0.9171	0.9500	0.0000	1.0000
dermatology	GaussianCopula	0.2222	0.1743	0.5017	249.0424	0.1643
dermatology	CTGAN	0.1806	0.1586	0.5002	262.4876	0.1560
dermatology	PATEGAN	0.2619	0.1910	0.5156	171.4425	0.2063
dermatology	PrivBayes	0.1389	0.1361	0.4851	237.2430	0.1903

5. Future Work

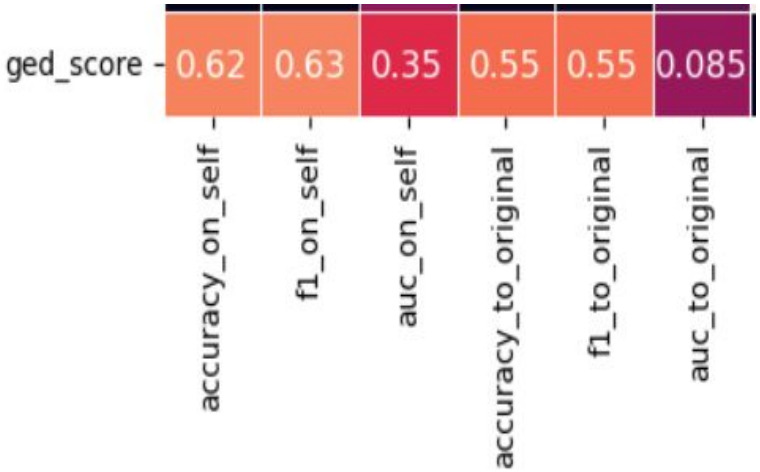
Proposed Method Benefits

Able to capture and measure utility using the model structure



*Can be **scaled for forest models**.

Tends to agree with contemporary utility metrics



*Important because **measures of structure usually do not correlate with model performance**, hence lack of structural measures being used for model utility.

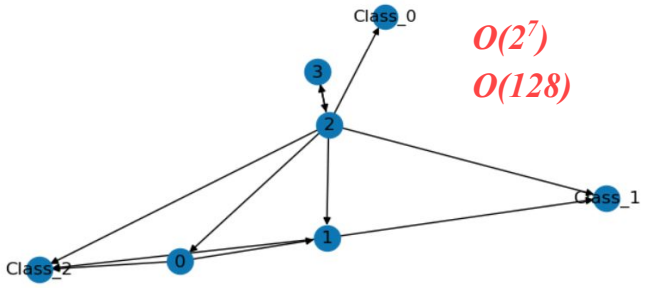
Limitations - Time Complexity

Exact GED is an NP-hard problem with current algorithms only making approximations.

Computation time increases significantly on datasets with more features/classes.

Datasets that have more features than observations are of concern and datasets with large amount of classes may also be of concern.

$O(2^n)$, $O(n^3)$, or $O(n^2 log(n))$ depending on approximation where n is the **amount of features and classes in the dataset.**



$O(2^7)$

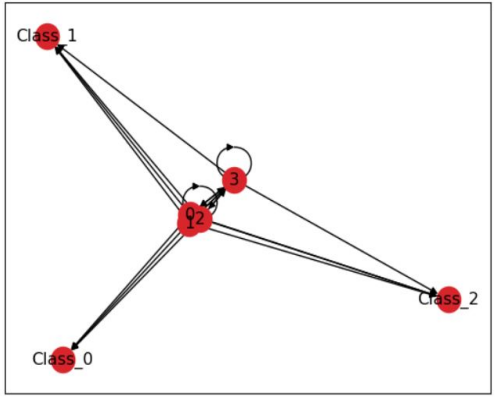
$O(128)$

Improvements - Tree to Graph

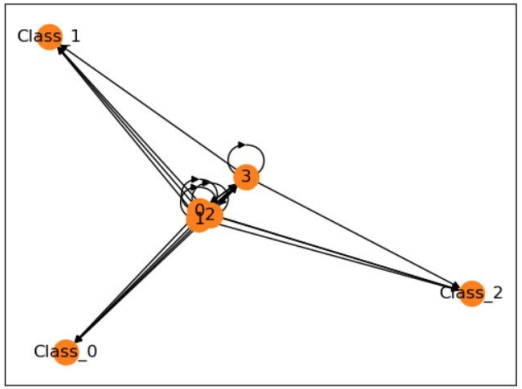
Differences between synthesizers are very small since most synthesizers produce similar graphs.

Room for improvement in the Tree to Graph protocol. Currently, we capture logical traversal and output traversal, but by potentially recording the average depth of a feature/class in a tree we could capture more structure information. Etc…

PATEGAN (Iris)



CTGAN (Iris)



Improvements - Algorithm Costs

Small differences between scores.

Room for improvement in the GED algorithm. Most of the difference in score is the result of the cosine similarity function which could be applied to the substitution function in the edges. Edge/node deletion and insertion could be tweaked too since synthetic and original graphs have the same features/classes.

Costs...

- Edge/Node Insertion: **1**
- Edge/Node Deletion: **1**
- Edge Substitution:
  - **0.5** if G_1[u][v].weight != G_2[u][v].weight
- Node Substitution:
  - **1 - cosine_sim**(G_1[x].avg, G_2[x].avg)

References

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Summer Research Synopsis

A Graph Edit Metric for Synthetic Data Utility

By Ethan Feldman under Dr. Lopresti Mentorship

NSF REU in Intelligent Scalable Systems. Lehigh Department of CS and Engineering